Susan Waithira Nyawira

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Professional Summary

Software developer and Computer Science graduate who builds software for real users and ships it. Hands-on with Java, Spring Boot, React, PostgreSQL and Docker, from backend APIs and databases through to deployment. Background in software engineering training, code review, ICT support and leading student project teams. I debug methodically, review code carefully and build software that the people running it day to day can trust.

Technical Skills

- **Languages:** Java, JavaScript, Elixir
- **Frameworks:** Spring Boot, React, Next.js, Node.js, Express, Phoenix, Tailwind CSS
- **Databases:** PostgreSQL, MySQL
- **Backend and APIs:** RESTful API design, API integration, JWT authentication, input validation, CRUD
- **DevOps and Tools:** Git, GitHub, Docker, Linux, Postman
- **Practices:** OOP, Data Structures and Algorithms, SDLC, Agile, automated testing, QA, requirements analysis (SSADM), networking fundamentals
- **Professional:** Debugging, code review, technical documentation, cross-team collaboration, team leadership

Projects

LendLedger: Loan Management Platform
Java, Spring Boot, React, PostgreSQL, Docker
- Full-stack loan management web application built from specification through to live deployment.
- REST API on a relational database, with business rule logic, an admin interface and automated tests.

Hesabu: Financial Systems Tool
Next.js, Tailwind CSS
- Built a web tool that calculates transaction fees, with the calculation logic kept separate from the user interface.
- Designed the interface for non-technical users, focusing on accessibility and clear results.

URL Shortener
Elixir, Erlang
- Built a URL shortening service that creates and resolves short links.

Authentication System
Node.js, JWT
- Secure user authentication with JWT sessions, password hashing and input validation.
- Tested endpoints for correct session handling.

Personal Portfolio Website
React, Tailwind CSS
- Responsive personal website with structured content, built for accessibility across devices. Full case studies and source links for each project are in the portfolio.

Work Experience

Software Engineering Trainee
Tracom Academy, Tracom Services Limited | Jun 2026 – Present
- Completed an intensive full-stack programme: Java (Spring Boot), Elixir (Phoenix), JavaScript, React, relational databases, Docker and DevOps, QA and testing.
- Designed, built and deployed a capstone web application end to end, from requirements to live deployment.
- Built RESTful APIs on relational databases and worked in Git-based team workflows.

ICT Support and Operations Attachment
CIC Insurance Group, BuruBuru Branch | 2025
- Supported daily business systems, including policy processing and client service workflows.
- Diagnosed and resolved hardware, software and connectivity issues to keep operations running.
- Worked with customer care, finance and CX teams to keep data handling accurate.

Coding Reviewer
African Centre for Women in ICT (ACWICT) | 2023 – 2024
- Reviewed codebases and gave structured feedback on readability, maintainability and best practices.
- Identified recurring issues and documented recommendations for the team.

Byte Scan Operator
Agricultural Society of Kenya | *2022 – 2023*
- Operated digital ticketing systems at high-volume public events, keeping transactions accurate.
- Diagnosed and resolved real-time system issues under pressure across multiple event cycles.

Education

Bachelor of Science in Computer Science, Technical University of Mombasa (Graduated 2026)
Coursework: Data Structures and Algorithms, Computer Networks, Database Systems, Software Engineering, Web Development

Leadership and Activities

- **Student Representative:** Represented student interests and coordinated academic engagement initiatives.
- **Team Manager, Group Projects:** Led student teams in delivering software projects, managing task allocation and timelines.

References

Available upon request